

Intelligent Farm Operations Platform with Smart Agro Precision System.

Farming : Crop Recommendations, Disease Detection,
Smart Irrigation, Satellite & Drone Monitoring
and Farmer Analytics.

Mentor
Ragul S

Team A Members

Aasima

Tejaswini

Anshul Chaurasiya

Ashutosh Mishra

Charitha S



Problem Statement

Challenges Faced by Farmers

1 No Real-Time Monitoring

Manual observation.
Delayed detection of crop stress.

2 Resource Wastage

Over-irrigation wastes 40-60% water.
Excess fertilizer degrades soil.

3 Unpredictable Weather

No early warning systems.
Crop failure from sudden changes.

4 Late Pest Detection

Identified too late.
₹90,000 Cr+ annual crop loss.

5 No Expert Access

Small farmers lack guidance.
Language barriers.

6 Fragmented Data

No centralized platform.
Decisions based on intuition.

OBJECTIVES -

FARMVERSE

- FarmVerse is a smart farming platform designed to support farmers in managing their farms more effectively.
- It helps farmers make informed decisions to improve crop growth and productivity.
- It promotes the efficient use of resources such as water, fertilizers, and land
- It assists in monitoring farm conditions and responding to changes in weather.
- It helps in detecting the disease early.
- Improves overall farm productivity while reducing operational cost.
- Support timely decisions for better farming outcomes.
- Encourage sustainable and environmentally friendly farming practices.
- Increase farmers' profitability by optimizing the use of available resources



FarmVerse

Precision Agriculture Management Platform — Modules



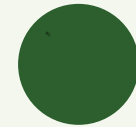
Farmer Register Module

faremer create an account



Farm Management Module

Add & manage fields, crops and daily farm activities



Sensor Monitoring Module

Live soil moisture, temperature , humidity ,soil pH, High Intensity, Water level



Weather Monitoring Module

Real-time forecasts, rainfall & climate risk updates



Smart Irrigation Module

scheduling to save water ,Prevents over-irrigation,reduce electricity cost



Crop Recommendation Module

based on soli pH,temparature,humidity,rainfall



Disease Detection Module

Image-based detection of crop diseases at early stage



Fertilizer Recommendation Module

Suggests right fertilizer type & dosage for each crop



Alert Module

Instant SMS/app alerts for risks, weather & sensor events



Dashboard Module

Unified view of farm health, analytics & reports

Unique Features

How FarmVerse Works Differently



Smart Dashboard

Map-based field viz
Real-time sensor overlay
Zone your farm into parcels

✦✦ Geospatial + temporal fusion



AI Crop Advisory

Upload leaf → instant detection
50K+ disease images trained
Works with smartphone

✦✦ Camera-based, no hardware



Smart Irrigation

Soil moisture + weather API
AI calculates exact water
Automated scheduling

✦✦ Saves 30-40% water

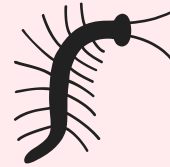
More Unique Features

Setting FarmVerse Apart



Weather

Micro-level predictions
Field-specific alerts
Frost/heatwave/rain warnings



Pest Prediction

7-14 day advance alerts
Weather + historical data
Preventive approach



Yield Prediction

ML-based forecasting
Historical trends
Market strategy planning



Marketplace

Buy seeds/fertilizers
Sell produce directly
Fair pricing



Voice Assistant

Local languages
Voice-based advisory
For non-tech farmers



Offline-First

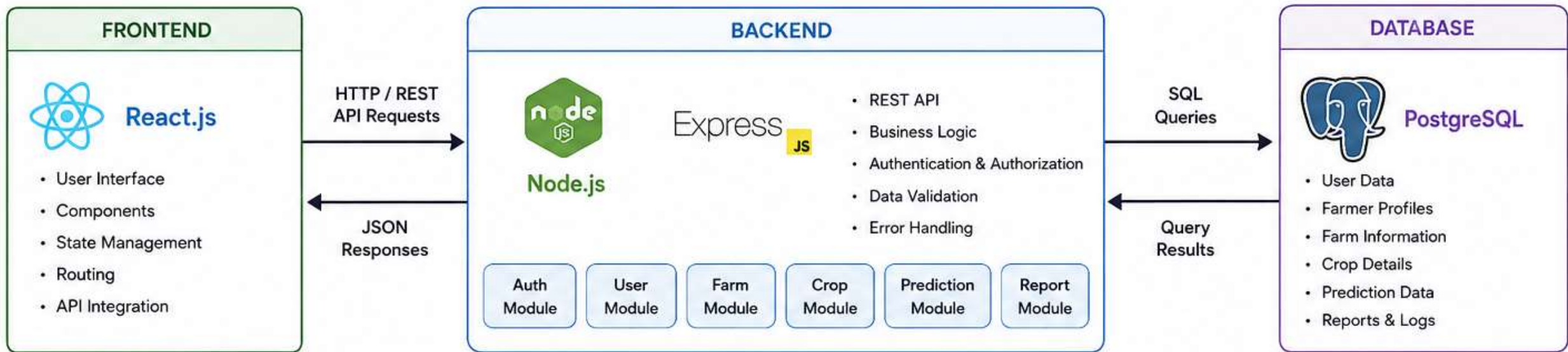
Works without internet
Syncs when connected
Rural India ready

? System Architecture

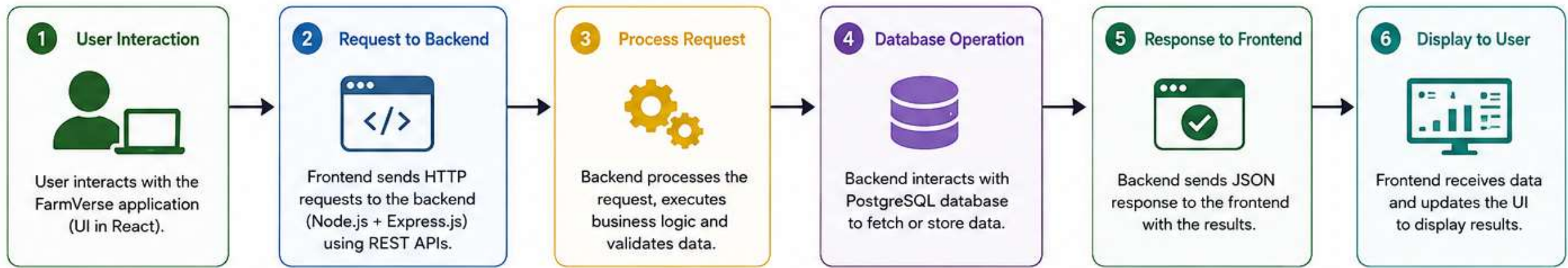
Smart, AI-Powered & Connected Farming Platform

FarmVerse – System Architecture

Smart, AI-Powered Agriculture Management Platform



WORKFLOW



Technology Stack

Tools & Technologies

FRONTEND

 React.js

 Tailwind CSS

BACKEND

 Node.js + Express

 Python FastAPI

 GraphQL + REST

 JWT Auth

DATABASE

 PostgreSQL

CLOUD & TOOLS

 GitHub Actions

 Vs code , IntelliJ IDEA

EXTERNAL APIs

 OpenWeatherMap

Key Features

Platform Capabilities

 Smart Map

 AI Advisor

 Irrigation

 Weather

 Pest Control

 Analytics

 Marketplace

 Voice Bot

 Community

 Mobile App

 Alerts

 Reports

✓ Benefits

Impact & Advantages



FOR FARMERS

📈 20-30% yield increase

💧 30-40% water savings

📦 25% less chemicals

💰 Better market prices



FOR ENVIRONMENT

📦 Reduced chemical runoff

💧 Water conservation

🌱 Sustainable farming

📦 Lower carbon footprint



FOR ECONOMY

📈 Higher profitability

🍏 Reduced food waste

🌐 Rural digitization

👤💻 Agri-tech jobs

Future Scope



Satellite Imagery

NDVI crop health from space
Large-scale monitoring
Detect stress before visible



Drone Monitoring

Automated aerial surveys
Precision spraying
3D terrain mapping



Advanced Analytics

ROI calculators
Carbon credit tracking
Water footprint analysis



Govt Integration

PM-KISAN link
Crop insurance
Subsidy recommendations



IoT Expansion

Smart greenhouses
Livestock monitoring
Aquaculture integration



Crop Genome DB

AI-recommended varieties
Soil-crop matching
Hybrid seed suggestions



Autonomous Farming

Robot harvesting
Self-driving tractors
Automated planting

Thank You